

Aviv Shaked

Cybersecurity Developer & Researcher · Haifa, Israel

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SUMMARY

C++20 developer and cyber researcher, currently a **Low-Level Android Developer** intern at a cyber intelligence company. Focused on network security, binary exploitation, and reverse engineering. Co-developed **HitchHiker**, a production covert communication platform that embeds encrypted messages into live network traffic with zero new packets and no observable footprint. Graduate of **Magshimim+**, Israel's elite national cybersecurity track, consistently self-driven beyond the curriculum.

TECHNICAL SKILLS

Languages	C++20 · Python · C · x86-64 Assembly · Java · C# · SQL · Bash
Cybersecurity	Binary exploitation · ROP chains · heap exploitation · format strings · ASLR bypass · network steganography · threat modeling · HackTheBox (RE, Pwn, Web, Crypto, Forensics)
Network Protocols	L2-L7 stack · TCP/IP · DNS · TLS · ICMP · ARP · DHCP · bit-level field manipulation · kernel packet interception (NFQueue) · iptables · Wireshark · Scapy
Reverse Engineering	Ghidra · IDA Pro · GDB · Frida · LIEF · Android & Java internals
Systems & Tooling	RAII · smart pointers · multithreading · socket programming (TCP/UDP/raw) · Linux internals (filesystem, permissions, syscalls) · CMake · Docker · Git · AWS EC2 · JetBrains
AI Agent Workflows	Claude Code · custom skills & plugins · multi-agent orchestration · MCP integration · hooks · context engineering · local LLMs (Ollama)
Cryptography	ECDH · ChaCha20 · AES · Argon2id · HKDF · SHA-256 · applied in production via Crypto++
AI / ML	Neural networks · minimax search · PyTorch · Scikit-learn · LLM/API integration · foundational

PROJECTS

HitchHiker: Network Steganography Platform C++20 · 2025-2026 · Completed

Co-developed with Daniel Varkin · Source available on request · Architecture: patcholie.github.io

C++20 Crypto++ NFQueue FTXUI Redis SQLite Docker AWS EC2 CMake

- Co-architected a covert communication platform that embeds encrypted messages into live network traffic: **zero new packets generated**, no observable network presence, undetectable by design
- Full cryptography stack (key exchange, symmetric encryption, key derivation, password hashing) applied at bit-field precision across multiple protocol layers via kernel-level packet interception
- Pluggable carrier architecture** behind a unified HitchHikerAPI; three independent CMake targets (lib / server / client), Redis + SQLite storage, full TUI (FTXUI); deployed on AWS EC2 via Docker

Multiplayer Trivia Game System C++ / C# / Unity · Magshimim · 2025

C++ C# Unity SQLite TCP Sockets

- Designed a custom application-layer protocol from scratch; concurrent TCP server, C# Unity client, and SQLite database integrated end-to-end with clean OOP architecture; scored 130/100 including bonus marks

Mancala-AI Python · PyTorch · 2025-2026

- Multi-agent AI system for Mancala with four agents (Random, Greedy, Heuristic, NN); NN uses 2-ply minimax with a trained MLP value function achieving 94% win rate vs Heuristic; full training pipeline, tournament evaluator, and GUI

Analisis: AI Wound Treatment Platform Python · 2022-2023 · First Project

- Co-led an AI-powered wound diagnosis and treatment platform; **won the national AI4Y competition**, then placed **2nd at the World Final** (2023)

EXPERIENCE

Low-Level Android Developer, Stealth May 2026 - Present · Internship · Hybrid

- Part-time internship at a cyber intelligence company; hands-on work in **low-level development**, **reverse engineering**, and complex systems, with AI integrated into the development workflow

EDUCATION

Magshimim Cyber, Advanced Track (Magshimim+) 2023-2026

Israel's elite national cybersecurity program, selective advanced track. Curriculum: networking, binary exploitation, reverse engineering, cryptography, systems programming (C/C++), Linux, team-based security development

BOSMAT, Haifa 2020-2026

Computer Science and AI, 10 study units (maximum track)

AWARDS & RECOGNITION

Magshimim+ Advanced Track · Graduate 2026 · Israel's elite national cyber program

Magshimim Cyber · Graduate 2026 AI4Y · National Winner, 2nd at World Final · 2023